

# WEB PROGRAMMING LAB-7 LAB EXERCISES

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**Name:** Devadathan N R

**Registration No:** 230905010

**Class:** CSE - A1

**Roll No:** 04

## Question 1

Develop a web application using Django framework to demonstrate the transfer of multiple parameters between web pages. User should be presented with a dropdown list containing car manufacturers, a text box which takes model name of the manufacturer and a submit button. On submitting the web page, the user is forwarded to a new page. This new page should display the selected car manufacturer name and the model name.

### Create project & App

```
# Create Django project
django-admin startproject form-project
cd form-project

# Create app
python manage.py startapp formapp
```

#### form-project/settings.py

```
import os # added at the top

INSTALLED_APPS = [
    'formapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

**Project URLs:** form-project/urls.py

```

from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('formapp.urls')),
]

```

#### App URLs: `formapp/urls.py`

```

from django.urls import path
from . import views

urlpatterns = [
    path('', views.index, name='index'),
    path('result/', views.result, name='result'),
]

```

#### View: `formapp/views.py`

```

from django.shortcuts import render

# Page 1
def index(request):
    return render(request, 'index.html')

# Page 2 (handles form submission)
def result(request):
    if request.method == "POST":
        manufacturer = request.POST.get('manufacturer')
        model = request.POST.get('model')

        error = None

        # VALIDATION
        if not model or model.strip() == "":
            error = "Model name cannot be empty"

        if error:
            return render(request, 'index.html', {'error': error})

    context = {
        'manufacturer': manufacturer,
        'model': model.strip()
    }

```

```
        return render(request, 'result.html', context)

    return render(request, 'index.html')
```

Create template/ in formapp/. Inside template/

**index.html**

```
<html>
<head>
    <title>Car Selection</title>
</head>
<body>

<h2>Select Car Details</h2>

{% if error %}
    <p style="color:red;">{{ error }}</p>
{% endif %}

<form action="{% url 'result' %}" method="POST">
    {% csrf_token %}

    <label>Choose Manufacturer:</label>
    <select name="manufacturer" required>
        <option value="">--Select--</option>
        <option value="Toyota">Toyota</option>
        <option value="Honda">Honda</option>
        <option value="BMW">BMW</option>
        <option value="Tesla">Tesla</option>
    </select>

    <br><br>

    <label>Enter Model:</label>
    <input type="text" name="model" placeholder="Enter model name" required>

    <br><br>

    <button type="submit">Submit</button>
</form>

</body>
</html>
```

**result.html**

```
<html>
<head>
```

```

<title>Result</title>
</head>
<body>

<h2>Selected Car Details</h2>

<p><strong>Manufacturer:</strong> {{ manufacturer }}</p>
<p><strong>Model:</strong> {{ model }}</p>

<br>

<a href="{% url 'index' %}">Go Back</a>

</body>
</html>

```

## Run

```
python manage.py runserver 8080
```

## Output

### index.html

### Select Car Details

Choose Manufacturer:

Enter Model:

### result.html

### Selected Car Details

**Manufacturer:** Toyota

**Model:** Innova

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## Question 2

Create a page firstPage.html with two TextBoxes [Name, Roll], DropDownList [Subjects], and a button. Create another page secondPage.html with a label and a

button. When the user clicks the button in first Page, he should be sent to the second page and display the contents passed from first page in the label. The button in second page should navigate the user back to firstPage. Use Django sessions to transfer information.

## Create project & App

```
# Create Django project
django-admin startproject session-project
cd session-project

# Create app
python manage.py startapp sessionapp
```

### session\_project/settings.py

```
import os # added at the top

INSTALLED_APPS = [
    'sessionapp', # added this line
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

## Database Initialization

To initialize the database and create required tables, executed:

```
python manage.py migrate
```

### Project URLs: session\_project/urls.py

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('sessionapp.urls')),
]
```

### App URLs: sessionapp/urls.py

```

from django.urls import path
from . import views
from django.urls import path
from . import views

urlpatterns = [
    path('', views.firstPage, name='first'),
    path('submit/', views.secondPage, name='submit'),
    path('back/', views.goBack, name='back'),
]
urlpatterns = [
    path('', views.cover, name='cover'),
]

```

View: `sessionapp/views.py`

```

from django.shortcuts import render, redirect

def firstPage(request):
    return render(request, 'firstPage.html')

def secondPage(request):
    if request.method == "POST":
        name = request.POST.get('name')
        roll = request.POST.get('roll')
        subject = request.POST.get('subject')

        # Validation
        if not name or not roll:
            return render(request, 'firstPage.html', {'error': 'All fields required'})

        if not roll.isdigit():
            return render(request, 'firstPage.html', {'error': 'Roll must be number'})

        # Store in session
        request.session['name'] = name
        request.session['roll'] = roll
        request.session['subject'] = subject

        # DIRECTLY render second page (no redirect)
        return render(request, 'secondPage.html', {
            'name': name,
            'roll': roll,
            'subject': subject,
        })

    return redirect('first')

```

```
def goBack(request):
    request.session.flush()
    return redirect('first')
```

Create template/ in sessionapp/. Inside template/

#### firstPage.html

```
<!DOCTYPE html>
<html>
<head>
    <title>First Page</title>
</head>
<body>

<h2>Enter Student Details</h2>

<!-- Error Message -->
{% if error %}
    <p style="color:red;">{{ error }}</p>
{% endif %}

<form action="{% url 'submit' %}" method="POST">
    {% csrf_token %}

    <label>Name:</label><br>
    <input type="text" name="name" required><br><br>

    <label>Roll:</label><br>
    <input type="text" name="roll" required><br><br>

    <label>Subject:</label><br>
    <select name="subject">
        <option value="Maths">Maths</option>
        <option value="Physics">Physics</option>
        <option value="CS">Computer Science</option>
    </select>

    <br><br>

    <button type="submit">Submit</button>
</form>

</body>
</html>
```

#### secondPage.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Second Page</title>
</head>
<body>

<h2>Submitted Details</h2>

<p><strong>Name:</strong> {{ name }}</p>
<p><strong>Roll:</strong> {{ roll }}</p>
<p><strong>Subject:</strong> {{ subject }}</p>

<br>

<form action="{% url 'back' %}">
    <button type="submit">Go Back</button>
</form>

</body>
</html>

```

## Run

```
python manage.py runserver 8080
```

## Output

### firstPage.html

### Enter Student Details

Name:

Roll:

Subject:

### secondPage.html



## Submitted Details

**Name:** Devadathan

**Roll:** 22

**Subject:** CS

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